

Project Report Format

1. INTRODUCTION

Project Overview:

The **Library Book Request Management System** is a ServiceNow application developed to automate the process of requesting and managing library books in an educational institution. The system provides a centralized platform where students can submit book requests online, while librarians can review, approve, reject, issue, and track requests efficiently. By replacing manual paper-based processes, the application improves transparency, reduces processing time, and ensures accurate record management.

Purpose.

1.2 Purpose

The purpose of this project is to simplify and digitize the library book request process using the ServiceNow platform. The application minimizes manual work, automates notifications and validations, and enables efficient management of library requests for both students and librarians.

2. IDEATIONPHASE

Problem Statement:

Traditional library book request systems rely on paper forms or email communication, resulting in delays, misplaced requests, and inefficient tracking. Students often cannot track the status of their requests, while librarians spend significant time managing requests manually. The proposed solution automates the complete request lifecycle using ServiceNow, improving efficiency, transparency, and user satisfaction.

Empathy Map Canvas:

Student

Says

- I want to request books quickly.
- I need to know my request status.

Thinks

- The manual process is slow.
- I hope my request reaches the librarian.

Does

- Fills out the request form.
- Waits for approval.

Feels

- Frustrated due to delays.
- Happy when requests are processed quickly.

Librarian

Says

- Managing requests manually is difficult.

Thinks

- There should be an automated system.

Does

- Reviews requests.
- Updates request status.

Feels

- Overloaded during busy periods.
- Satisfied when work is automated.

Brainstorming:

Ideas generated during brainstorming include:

- Online Book Request Form
- Auto-generated Request Number
- Issue Date Validation
- Email Notification
- Status Tracking
- Flow Designer Automation
- Book Issue & Return Management
- Student Information Management
- Librarian Dashboard

The team selected **Library Book Request Management System** because it effectively addresses manual library operations using ServiceNow automation.

3. REQUIREMENT ANALYSIS

Customer Journey map:

- Student opens the application.
- Student fills in request details.
- Student submits the request.
- System validates issue date.
- Request number is generated automatically.
- Email notification is sent.
- Librarian reviews the request.
- Librarian approves or rejects the request.
 - Student receives the updated request status

SolutionRequirement:

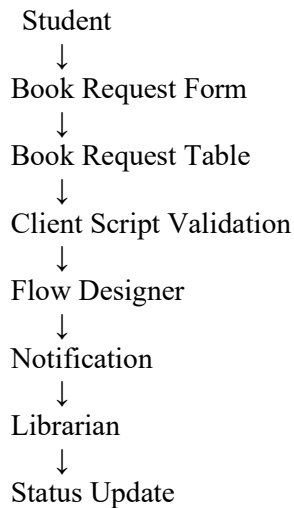
Functional Requirements

- Student registration of book requests
- Auto-generated request number
- Book details management
- Issue and return date management
- Status tracking
- Email notification
- Client-side validation
- Flow Designer automation

Non-Functional Requirements

- Secure access
- Fast response time
- User-friendly interface
- High availability
- Data accuracy

Data Flow Diagram:



Technology Stack:

Component

Platform

Development

Database

Automation

Validation

Notification

Language

Technology

ServiceNow

App Engine Studio

ServiceNow Tables

Flow Designer

Client Scripts

Email Notifications

JavaScript

11. PROJECT DESIGN

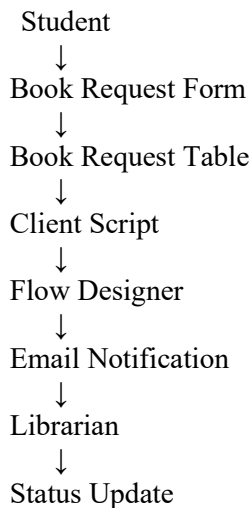
Problem Solution Fit:

The proposed solution directly addresses the challenges of manual library management by automating request submission, approval, validation, and notification processes. Students receive faster responses while librarians can manage requests more efficiently.

Proposed Solution:

Develop a ServiceNow-based application that allows students to submit library book requests online. The application validates user inputs, generates request numbers automatically, sends notifications to librarians, and maintains complete request records with status updates.

Solution Architecture:



12. PROJECT PLANNING & SCHEDULING

Project Planning:

Phase	Activity
Week 1	Requirement Analysis
Week 2	Application Creation
Week 3	Table & Field Creation
Week 4	Client Script Development
Week 5	Notification & Flow Designer
Week 6	Testing & Documentation

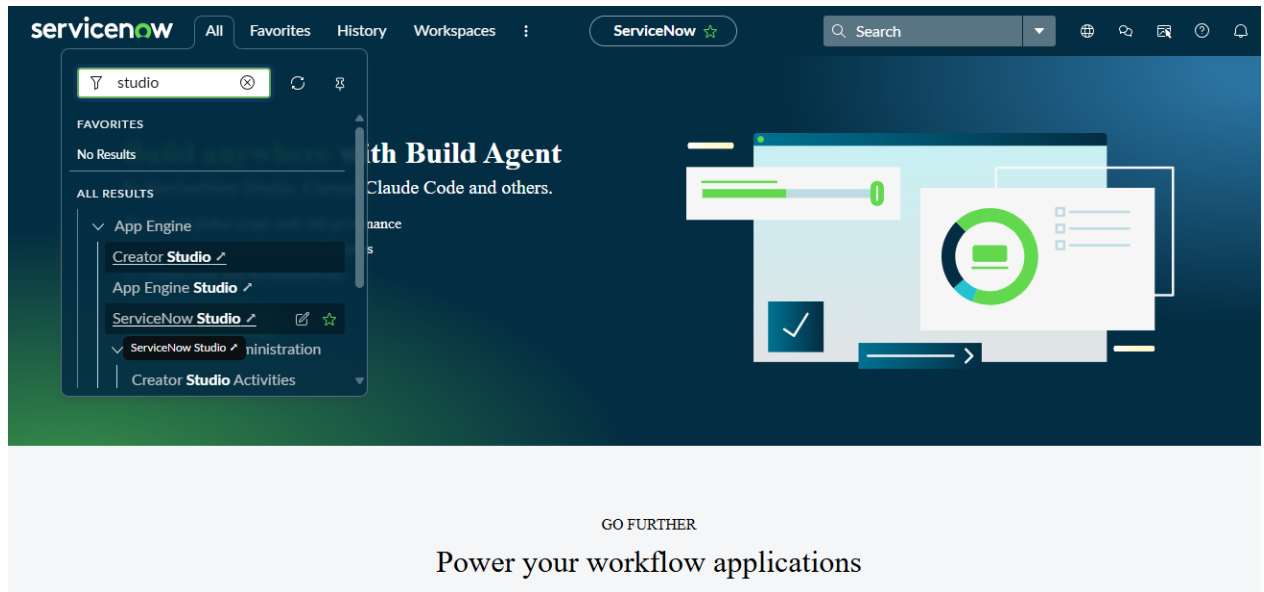
19. FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing:

Test Case	Expected Result	Status
Create Book Request	Record Created	Pass
Auto Number Generation	Unique Number Generated	Pass
Issue Date Validation	Past Date Rejected	Pass
Email Notification	Notification Sent	Pass
Flow Execution	Successfully Executed	Pass
Status Update	Updated Successfully	Pass

20. RESULTS

Application Home



Book Request Form

The screenshot shows the ServiceNow Book Request form. The top navigation bar includes the ServiceNow logo, tabs for All, Favorites, History, and Workspaces, a search bar, and a user profile icon. The form title is 'Book Request - Create BOO0001004'. The form fields are as follows:

Field	Value
Number	BOO0001004
Student Name	
Student ID	
Book Name	
Author	
Issued Date	
Return Date	
Phone Number	
Email	

A 'Submit' button is located at the bottom right of the form.

Client Script

servicenow All Favorites History Admin : Client Script - Validate Issue Date Search

Client Script
Validate Issue Date

Script

```

1 function onSubmit() {
2
3     var issueDate = g_form.getValue('u_issue_date');
4
5     if (!issueDate)
6         return true;
7
8     var today = new Date();
9     today.setHours(0, 0, 0, 0);
10
11     var selectedDate = new Date(issueDate);
12     selectedDate.setHours(0, 0, 0, 0);
13
14     if (selectedDate < today) {
15         alert("Issue Date cannot be in the past.");
16         return false;
17     }
18
19     return true;
20 }

```

Isolate script ☒

Update Delete

Notification

servicenow All Favorites History Admin : Notification - New Record Search

Notification [sysevent_email_action]
New record

- Who receives the notification
- What content is in the notification

Name Application ⓘ

* Table Active ☒

Mandatory ☐

* Category ⓘ

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest ☐

When to send Who will receive What it will contain

Notifications can be sent (if the specified **Conditions** are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Flow Designer

Workflow Studio
New Flow
Flow • None

Let's get the details for your flow

Flow name * ⓘ

Application * ⓘ

Description ⓘ

[Show additional properties](#)

Cancel
Build flow

Auto-generated Number

servicenow
All
Favorites
History
Book Request - Create BOO0001004
Search

Book Request
New record
Submit

Number
BOO0001004

Student Name

Student ID

Book Name

Author

Issued Date

Return Date

Phone Number

Email

Submit

If issued date is past:

The screenshot shows the 'Book Request' form in ServiceNow. A validation error message is displayed: 'dev265942.service-now.com says Issue Date cannot be in the past.' The form fields are as follows:

Number	BOO0001013
Student Name	Devaki
Student ID	23a51a0527
Book Name	ServiceNow
Author	mohandas
Issued Date	2026-06-28
Return Date	2026-07-02
Status	Requested
Phone Number	1234565432
Email	kandalamdevaki1316@gmail.com

A 'Submit' button is visible at the bottom left of the form.

Email Preview

The screenshot shows the 'Email - New Book Request Submitted' page in ServiceNow. A 'Preview Email' dialog box is open, displaying the following content:

Preview Email

Hello Librarian,
A new library book request has been submitted.
Please check the Book Request record for the request details.
Thank you.

Ref:MSG0001801_azsqIUBGgONATmmG9fHq

The background shows the email record details, including 'Content type' and 'Headers'. Below the preview, there are 'Update' and 'Delete' buttons, and a 'Related Links' section with a 'Preview Email' link. At the bottom, there is an 'Email Log' section with a search bar and a table with columns: Created, Level, Message, and Source.

21. ADVANTAGES&DISADVANTAGES

Advantages

- Eliminates manual paperwork
- Faster request processing
- Improved transparency
- Easy request tracking
- Automatic notifications
- Better record management
- User-friendly interface

- Improved efficiency

Disadvantages

- Requires ServiceNow platform access
- Email delivery depends on instance configuration
- Internet connection is required
- Initial configuration requires administrator access

22. CONCLUSION

The **Library Book Request Management System** successfully automates the library request process using ServiceNow. It provides students with an easy way to request books while enabling librarians to manage requests efficiently. Automation through client scripts, notifications, and Flow Designer improves accuracy, reduces manual effort, and enhances the overall user experience.

23. FUTURE SCOPE

- Book availability checking
- Barcode/QR code integration
- Fine calculation for overdue books
- Student dashboard
- Librarian dashboard
- Reports and analytics
- Mobile application support
- SMS notifications
- Integration with library inventory systems
- AI-based book recommendations

24. APPENDIX

SourceCode(ifany) Dataset

Link

GitHub&ProjectDemo Link